

Derivative Chart Practice - Product and Quotient Rule

Directions: Use each table to find the indicated derivative. Show all work and simplify your answer.

1. $h(x) = f(x)g(x)$, find $h'(2)$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	-2	3	4	-1
2	5	-2	1	3
3	1	4	-3	2
6	-4	5	2	-2

2. $h(x) = \frac{f(x)}{g(x)}$, find $h'(3)$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	6	-3	2	1
2	-2	4	-1	3
3	5	2	4	-2
6	1	-5	2	1

3. $h(x) = f(x)g(x) + x^2$, find $h'(1)$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	4	-2	3	5
2	-3	6	2	-1
3	5	1	-2	4
6	2	-4	1	3

4. $h(x) = \frac{f(x)}{g(x)} - 2x$, find $h'(6)$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	-5	2	1	4
2	3	-3	-2	1
3	4	5	3	-2
6	-2	4	5	1

5. $h(x) = 3f(x)g(x)$, find $h'(3)$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	2	-4	-3	2
2	-1	5	4	-1
3	6	2	1	3
6	3	-5	-2	4

6. $h(x) = \frac{2f(x)}{g(x)}$, find $h'(1)$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	3	2	-4	5
2	6	-3	2	1
3	-2	4	3	-1
6	5	1	-2	4

7. $h(x) = f(x)g(x) - x^3$, find $h'(6)$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	-4	3	2	-2
2	1	-5	4	3
3	6	2	-1	5
6	2	4	3	-1

8. $h(x) = \frac{f(x)}{g(x)} + x^2 - 4x$, find $h'(2)$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	5	-2	1	4
2	-3	5	2	-2
3	4	1	-3	2
6	6	-4	1	3

9. $h(x) = 2f(x)g(x) + 3x^2$, find $h'(1)$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	2	3	-1	4
2	-5	1	3	-2
3	4	-4	2	1
6	1	5	-3	2

10. $h(x) = \frac{3f(x)}{2g(x)} - x^2$, find $h'(3)$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	-1	4	2	3
2	5	-2	-3	1
3	2	5	4	-1
6	-4	3	1	2

11. $h(x) = f(x)g(x) + 4x^3 - x$, find $h'(2)$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	6	-3	2	-1
2	-2	4	5	3
3	1	2	-4	1
6	3	-5	2	4

12. $h(x) = \frac{2f(x)}{g(x)} + x^3 - 3x^2$, find $h'(6)$

x	$f(x)$	$g(x)$	$f'(x)$	$g'(x)$
1	4	5	-2	1
2	-3	2	4	-1
3	6	-4	1	3
6	2	3	-5	2